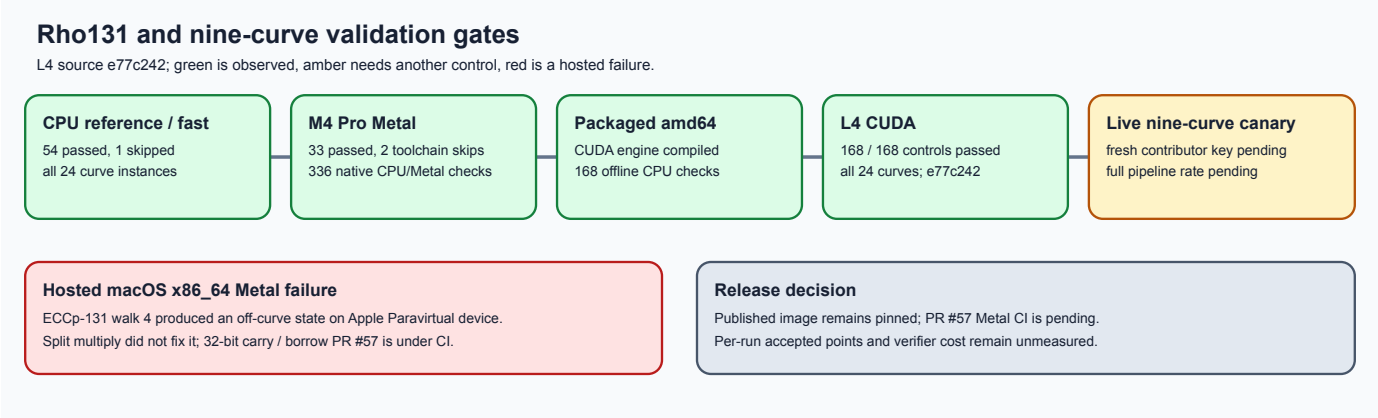
Rho131 current-head and nine-curve validation

Snapshot: 2026-10-24. Base source commit `b78fc2921e9b9a60174e1693aff15c7daf23f9c7`. The public catalog currently pins `sha256:a347a784935a7713b204335e8316f780a5c3209d64cc372980c1b35bee9f81ba`; that image predates this base commit. The report is an execution snapshot, not a release or a full-campaign performance claim.



Observed controls

On the Apple M4 Pro, the base source passed 54 CPU tests with one skip. The Metal parity selection passed 33 tests with two offline-toolchain skips. The native control runner passed 336 CPU/Metal checks across all 24 exact instances, including all nine requested binary curves. A local amd64 image built from the base source compiled and passed its native self-test; its packaged, network-disabled CPU check passed 168 controls. These are distinct from L4 CUDA execution of the new source.

The first local Docker build mixed an arm64 engine build stage with an amd64 runtime stage. Its offline check failed to start the engine. An explicit `linux/amd64` build corrected the executable architecture, then the packaged check passed. Both build logs are retained. This was a build invocation error; the corrected image has not been published.

The current-head GitHub worker-platform run reported one failing case: ECCp-131 walk 4 on the macOS x86_64 Apple Paravirtual Metal device produced an off-curve state at five steps. The same targeted case passed with an x86_64 process under Rosetta on the local M4 Pro. A diagnostic rerun on the hosted Apple Paravirtual device failed with the same case and state. That distinguishes the two observed devices but does not establish the cause. A 32-bit-only Metal multiplication path and a repeated parity control were then tested on hosted Apple Paravirtual in PR 55, run 40070444197. The same invalid five-step state recurred; the split product did not repair the failure. PR 57 changes the prime GPU limb carry and borrow operations to 32-bit arithmetic. Its local boundary tests and CPU controls passed, while hosted Apple Paravirtual CI remains pending. The cause has not been established.

L4 CUDA control on the repair commit

Commit `e77c24294aaac8a70bddbb219a6a572650e5fdfc` ran the key-free current-source L4 control against the archived CPU inputs. The NVIDIA L4 engine binary SHA-256 was `2ec2935e3aac237986711387371eb4d5c09225757a41220a8751bec4c2cbd871`. The verifier checked all 168 returned case hashes across all 24 curves, including the nine requested binary curves, against the archive; 168 passed. The source, inputs, binary, device and per-case hashes are in `l4-control-e77c242/`. This establishes bounded CUDA byte parity; it does not measure sustained worker rate or gateway/verifier cost.

Exploratory matched invocation timing

The frozen walk-2 ECC2-131 inputs were run on the base source with bracketed reference and specialised engine calls, plus a reference A/A control. All compared output bytes and actual step counts matched. Raw timings and ratios are retained in `local/matched-cpu/` and `local/matched-metal/` as exploratory observations on a shared host. They exclude registration, network, distinguished-point upload, server verification and provider billing. No performance advantage is promoted from these rows. The proposed Metal source was also measured on the native M4 Pro path with exact output and useful-step equality. Its Apple Paravirtual variant remains unmeasured.

Open gates

PR 57 hosted Apple Paravirtual CI remains open. The authenticated nine-curve canary needs the fresh dedicated contributor key stored in a local mode-0600 file. Its planned launcher uses one L4 sandbox at a time, a fresh volume per curve, immutable image identity, authenticated per-run status, checkpoint restart, accepted records, index completion and closure. A no-point attempt is retained as a censored result. Full contributor and verifier cost comparisons remain open until those runs produce bound receipts.

The artifact index is `README.md`; the hosted failure is GitHub Actions run 40066130516, job 120789662742. The editable diagram is `gates.svg`.